

Semester Test 3

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Memo

Electricity and Electronics EBN 122

24 October 2019

Test Information

Maximum marks:	46	Full marks:	45
Duration of test:	90 min + (10 min for OCR)	Open/Closed book:	Closed
Total number of pages (this page included)		18	

Important

1. The examination regulations of the University of Pretoria apply.
2. Answer all questions on the OCR forms provided. The question number 01 in this question paper refers to the numeric field 01 of the OCR form.
3. Read the instructions on the OCR form carefully.
4. Where applicable, answers must include units.
5. Final answers must be written as real numbers and not as fractions.
6. Use at least 3 significant numbers to write irrational numbers.

Lecturers: Prof. J.P. de Villiers, Prof. J.P. Jacobs, Dr X. Ye

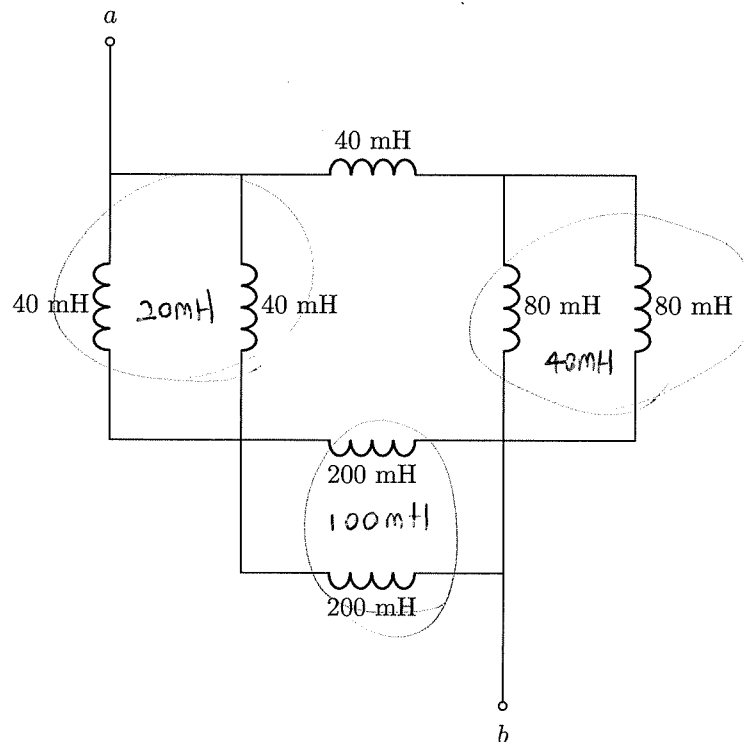
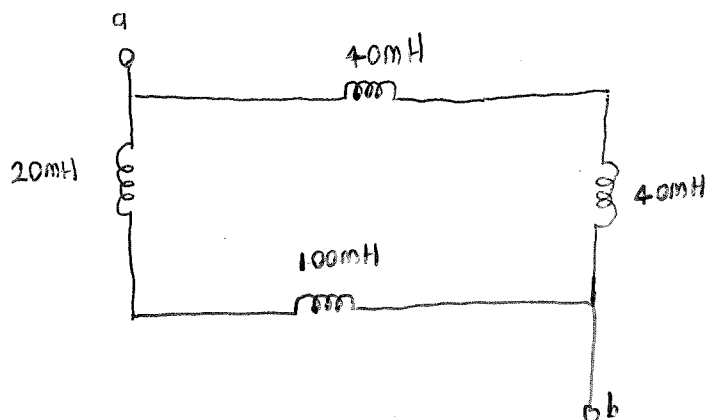
INSTRUCTIONS**Do step 0 in the first 5 minutes of the session. (-5 to 0 min)****Do step 1 in the first 30 minutes of the test. (0-90 min)****Do step 2 within the following 10 minutes. (90-100 min)****STEP 0 (-5 min to 0 min):** Complete the front page of the OCR form.**STEP 1 (0 min to 90 min):** Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.**STEP 2 (90 min to 100 min):** (only if numeric sheets are completed) Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

- The Assessment ID is: **2019EBN122S03**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1

Question 1 [2]

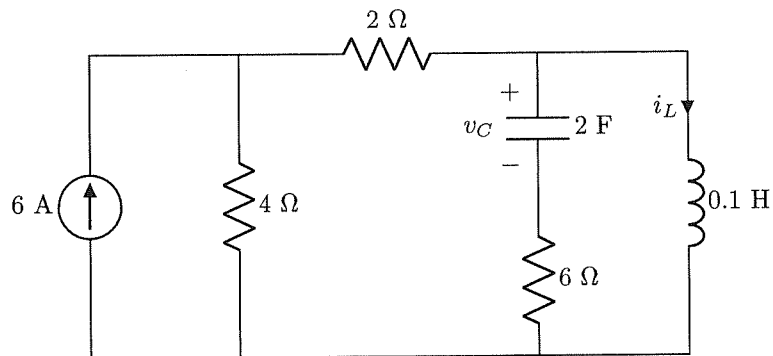
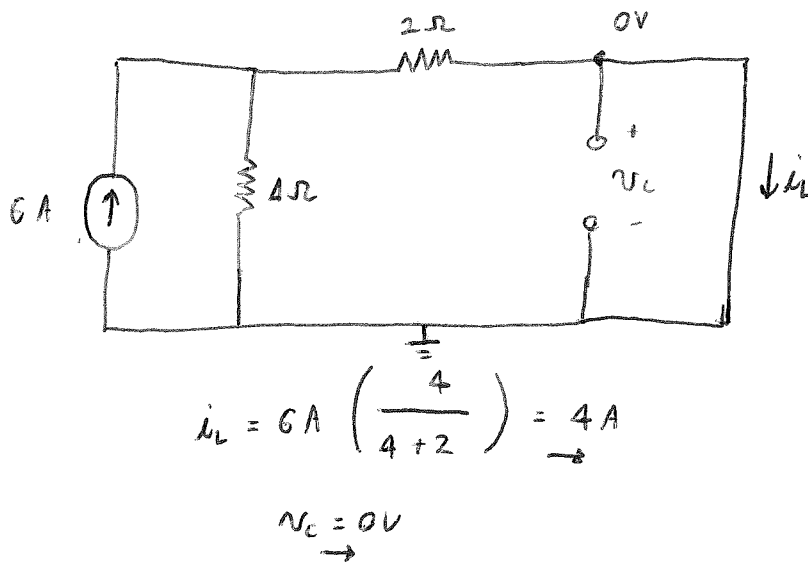
Consider the circuit below.

01 Find L_{eq} when looking into the terminals $a - b$. [2]

$$\begin{aligned}
 L_{eq} &= (20\text{mH} + 100\text{mH} // 40\text{mH} + 40\text{mH}) \\
 &= (120\text{mH} // 80\text{mH}) \\
 &= 48\text{mH}
 \end{aligned}$$

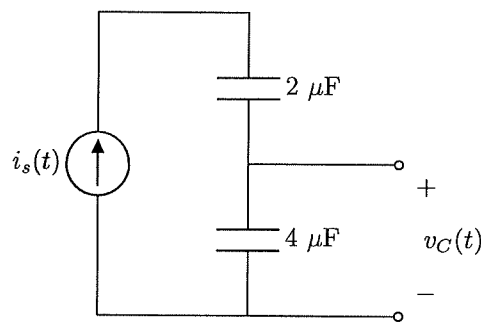
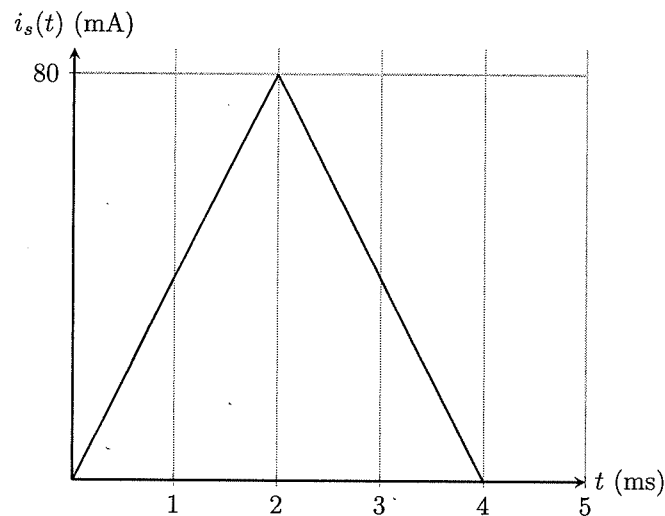
Questions 2 to 3 [2]

Consider the circuit below, and assume steady state DC conditions.

02 Find v_C . [1]03 Find i_L . [1]

Questions 4 to 5 [2]

The current $i_s(t)$ charges the two capacitors, which are initially uncharged.



Find $v_C(t)$ at $t = 1$ ms and $t = 3$ ms.

04 $v_C(t = 1 \text{ ms})$ [1]

05 $v_C(t = 3 \text{ ms})$ [1]

$$\begin{aligned} \underline{04}: v_C(1) &= \frac{1}{C} \int_0^1 i_s(t) dt \\ &= \frac{1}{4 \times 10^{-6}} \left[\frac{1}{2} (10^{-3}) (40 \times 10^{-3}) \right] \\ &= 5 \text{ V} \end{aligned}$$

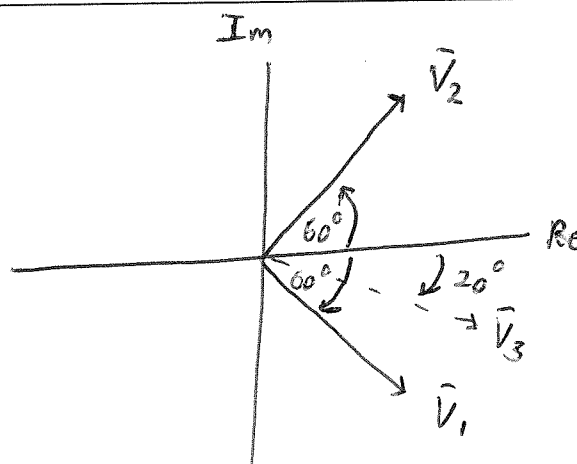
$$\begin{aligned} \underline{05}: v_C(3) &= v_C(1) + \frac{1}{C} \int_1^3 i_s(t) dt \\ &= \frac{1}{4 \times 10^{-6}} \left[\frac{1}{2} (2 \times 10^{-3}) (80 \times 10^{-3}) \times 2 - \frac{1}{2} (10^{-3}) (40 \times 10^{-3}) \right] \\ &= 40 - 5 \\ &= 35 \text{ V} \end{aligned}$$

Questions 6 to 8 [4]	
Consider three sinusoid waveforms $v_1(t)$, $v_2(t)$, and $v_3(t)$, where $v_1(t) = 12 \sin(\omega t + 30^\circ), \text{ and}$ $v_2(t) = 5 \cos(\omega t + 60^\circ).$ The waveforms $v_1(t)$, $v_2(t)$, and $v_3(t)$ can be denoted by phasors $\mathbf{V}_1$, $\mathbf{V}_2$, and $\mathbf{V}_3$ respectively. The phase angle between $\mathbf{V}_1$ and $\mathbf{V}_3$ is ϕ_0, and the phase angle between $\mathbf{V}_2$ and $\mathbf{V}_3$ is $2\phi_0$; $\mathbf{V}_3$ is in the 4th quadrant of the phasor domain.	
06	Choose the correct statement. ① v_2 leads v_3. 2. v_1 leads v_3. 3. v_1 leads v_2. Write 1 if statement 1 is correct, 2 if statement 2 is correct, or 3 if statement 3 is correct. [2]
07	Find $\phi_0 \in [0^\circ, 180^\circ]$. [1]
08	Use phasors to find $\mathbf{V} = \mathbf{V}_1 + \mathbf{V}_2$ in polar form. [1]

06: $v_1(t) = 12 \cos(\omega t - 60^\circ)$

$$\bar{V}_1 = 12 \angle -60^\circ$$

$$\bar{V}_2 = 5 \angle 60^\circ$$



07: $\phi_{\bar{V}_1} = -60^\circ$; $\phi_{\bar{V}_2} = 60^\circ$

$$\phi_{\bar{V}_2} - \phi_{\bar{V}_1} = 120^\circ = 2\phi_0 + \phi_0$$

$$\phi_0 = 40^\circ$$

$\bar{V}_2$ leads $\bar{V}_3$

08: $\bar{V} = \bar{V}_1 + \bar{V}_2$

$$= 12 \angle -60^\circ + 5 \angle 60^\circ$$

$$= 10,44 \angle -35,50^\circ \text{ V}$$

→

Questions 9 to 10 [4]	
Simplify the following expression and give the answer in rectangular form:	
$Z = \frac{(4 + j3)^2}{3 \angle 20^\circ} (16 - j20)^*$	
where * refers to the complex conjugate.	
09	Z [2]
Find I_2 in polar form.	
$\begin{bmatrix} 4 \angle 60^\circ & 3 - j4 \\ -2 & 8 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} -8 \angle 30^\circ \\ 0 \end{bmatrix}$	
10	I_2 [2]

09 :

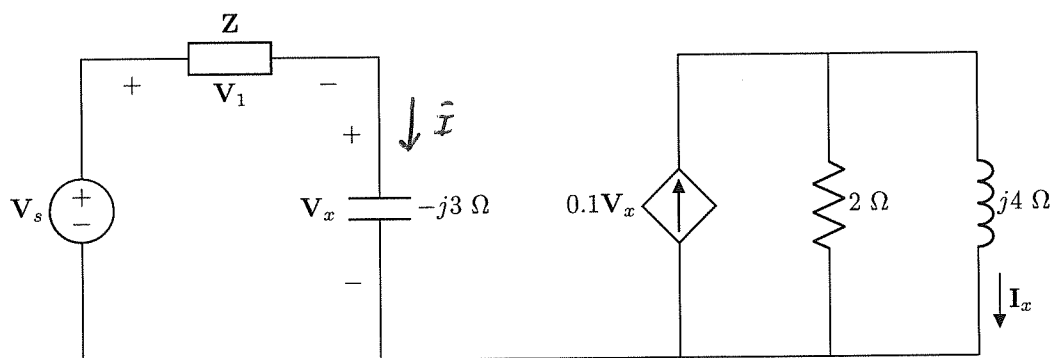
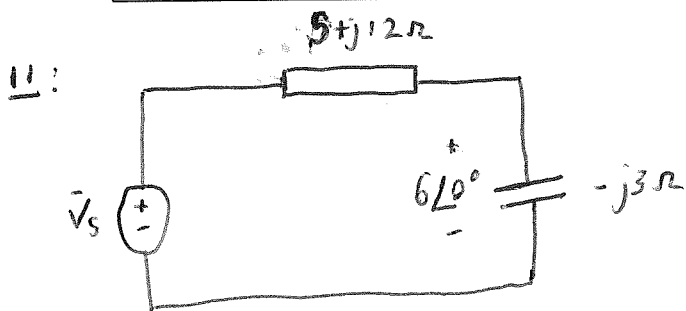
$$\begin{aligned}
 \bar{Z} &= \frac{(4 + j3)^2}{3 \angle 20^\circ} (16 - j20)^* = \frac{7 + j24}{3 \angle 20^\circ} (16 + j20) \\
 &= \frac{25 \angle 73,74^\circ}{3 \angle 20^\circ} (25,61 \angle 51,34^\circ) \\
 &= -55,5 + j206,07 \\
 &\quad \rightarrow
 \end{aligned}$$

10 :

$$\begin{aligned}
 \bar{I}_2 &= \frac{\begin{vmatrix} 4 \angle 60^\circ & -8 \angle 30^\circ \\ -2 & 0 \end{vmatrix}}{\begin{vmatrix} 4 \angle 60^\circ & 3 - j4 \\ -2 & 8 \end{vmatrix}} = \frac{(8 \angle 30^\circ)(-2)}{(4 \angle 60^\circ)(8) - (3 - j4)(-2)} \\
 &= 0,541 \angle 168,14^\circ \\
 &\quad \rightarrow
 \end{aligned}$$

Questions 11 to 12 [4]

Consider the circuit below.

11 Given $Z = 5 + j12 \Omega$ and $V_x = 6 \text{ V}$, find V_s in polar form. [2]12 If $I_x = 2 \angle 0^\circ \text{ A}$ and $V_1 = -1.4 + j4.8 \text{ V}$, then find Z in rectangular form. [2]

KVL: $-\bar{V}_s + (-24 + j10) + 6 \angle 0^\circ = 0$
 $\bar{V}_s = 20.59 \angle 150.95^\circ \text{ V}$

12:

$$\bar{I}_x = 0.1 \bar{V}_x \left[\frac{2}{2 + j4} \right] = 2 \angle 0^\circ$$

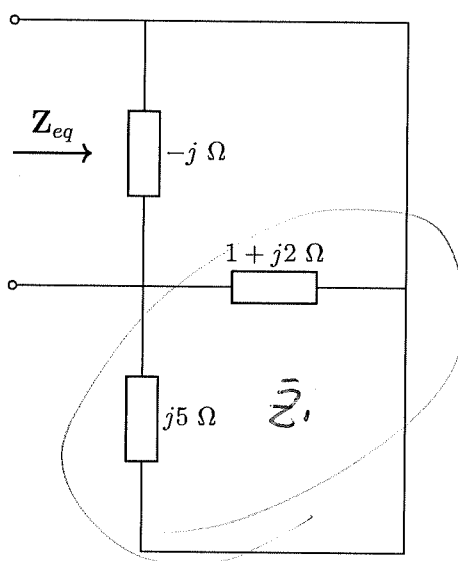
$$\bar{V}_x = 20 + j40 \text{ V}$$

$$\bar{I} = \frac{20 + j40}{-j3} = 14.91 \angle 153.43^\circ \text{ A}$$

$$\bar{Z} = \frac{\bar{V}_1}{\bar{I}} = \frac{-1.4 + j4.8}{14.91 \angle 153.43^\circ} = 0.228 - j0.246 \Omega$$

Question 13 [2]

Consider the circuit below.

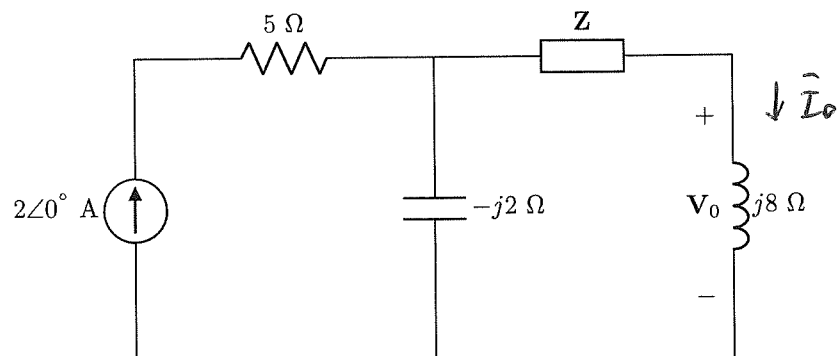
13 Find Z_{eq} in rectangular form. [2]

$$\bar{Z}_1 = \left(\frac{1}{j5} + \frac{1}{1+j2} \right)^{-1} = 0,5 + j1,5 \Omega$$

$$\bar{Z}_{eq} = \left(\frac{1}{-j} + \frac{1}{0,5 + j1,5} \right)^{-1} = 1 - j2 \Omega$$

Question 14 [2]

Consider the circuit below.

14 Find Z when $V_0 = 4\angle 0^\circ \text{ V}$ in rectangular form. [2]

$$\underline{14:} \quad 2\angle 0^\circ \left(\frac{-j2}{\bar{Z} + j8 - j2} \right) = -j0,5 \text{ A} \quad \rightarrow \quad \bar{I}_0 = \frac{\bar{V}_0}{j8} = -j0,5 \text{ A}$$

$$\cancel{14} \quad -j2 = -j0,25 (\bar{Z} + j6)$$

$$\bar{Z} = 8 - j6 \Omega$$

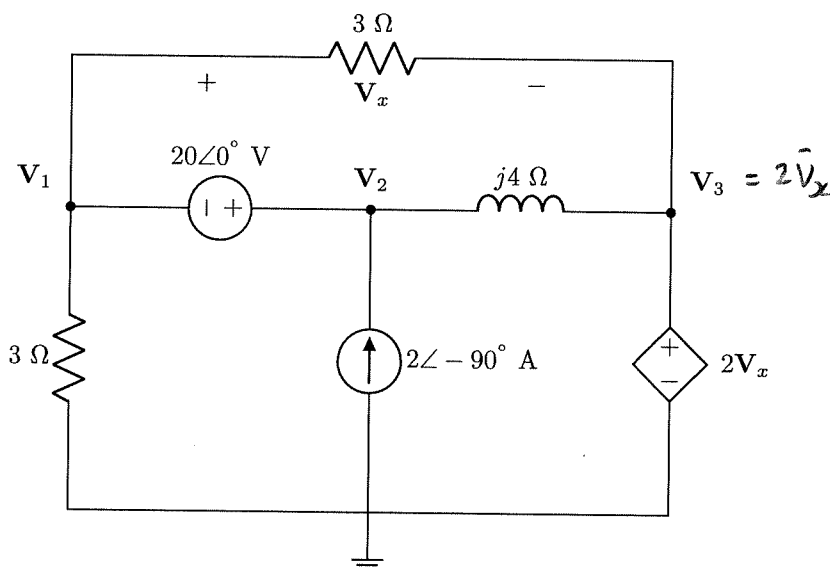
Questions 15 to 18 [4]

Use nodal analysis to set up two equations of the following form:

$$aV_1 + bV_2 = -20\angle 0^\circ,$$

$$cV_1 + 9V_2 = d.$$

Find the complex values **a**, **b**, **c**, and **d** in **rectangular** form.



15	Determine a. [1]
16	Determine b. [1]
17	Determine c. [1]
18	Determine d. [1]

$$\bar{V}_1 + 20\angle 0^\circ = \bar{V}_2$$

$$\boxed{\bar{V}_1 - \bar{V}_2 = -20\angle 0^\circ} \rightarrow \boxed{\bar{a} = 1; \bar{b} = -1}$$

$$\frac{\bar{V}_1 - 2(\bar{V}_1/3)}{3} + \frac{\bar{V}_1}{3} - 2\angle -90^\circ + \frac{\bar{V}_2 - \frac{2\bar{V}_1}{3}}{j4} = 0$$

$$\bar{V}_1 \left[\frac{1 - \frac{2}{3}}{3} + \frac{1}{3} - \frac{2}{j12} \right] + \bar{V}_2 \left[\frac{1}{j4} \right] = 2\angle -90^\circ$$

$$\bar{V}_1 \left[\frac{4}{9} + \frac{j}{6} \right] + \bar{V}_2 \left[\frac{1}{j4} \right] = 2\angle -90^\circ \quad (\times j36)$$

$$\bar{V}_1 [-6 + j16] + \bar{V}_2 [9] = 72$$

$$\boxed{\bar{c} = -6 + j16; \bar{d} = 72}$$

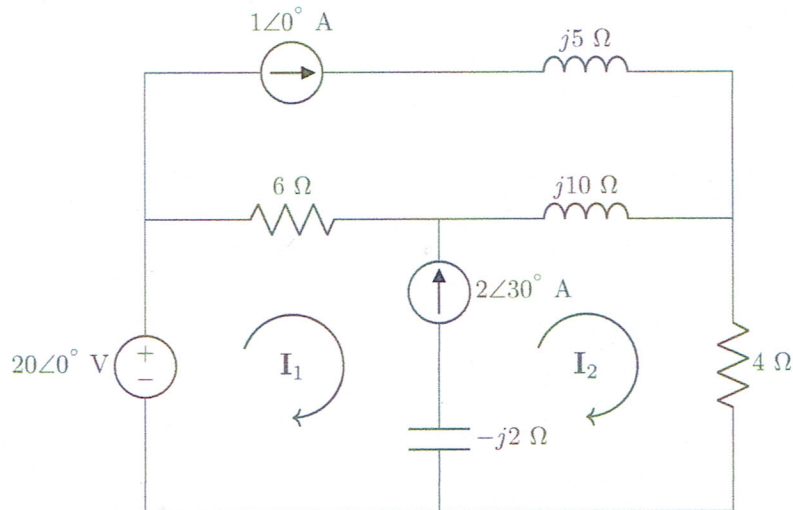
Questions 19 to 22 [4]

Use mesh analysis to set up two equations of the following form:

$$\mathbf{I}_1 + \mathbf{e}\mathbf{I}_2 = \mathbf{f},$$

$$6\mathbf{I}_1 + \mathbf{g}\mathbf{I}_2 = \mathbf{h}.$$

Find the complex values $\mathbf{e}$, $\mathbf{f}$, $\mathbf{g}$, and $\mathbf{h}$ in rectangular form.



19	Determine $\mathbf{e}$. [1]
20	Determine $\mathbf{f}$. [1]
21	Determine $\mathbf{g}$. [1]
22	Determine $\mathbf{h}$. [1]

$$\bar{\mathbf{I}}_1 - \bar{\mathbf{I}}_2 = -2\angle 30^\circ$$

$$\bar{\mathbf{e}} = -1 ; \bar{\mathbf{f}} = 2\angle -150^\circ = -1,73 - j$$

Supermesh: $-20\angle 0^\circ + 6(\bar{\mathbf{I}}_1 - 1) + j10(\bar{\mathbf{I}}_2 - 1) + 4\bar{\mathbf{I}}_2 = 0$

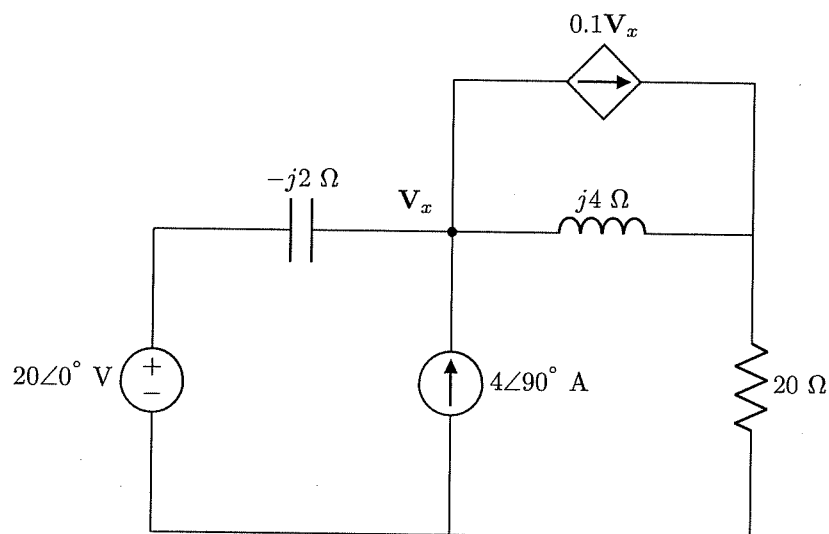
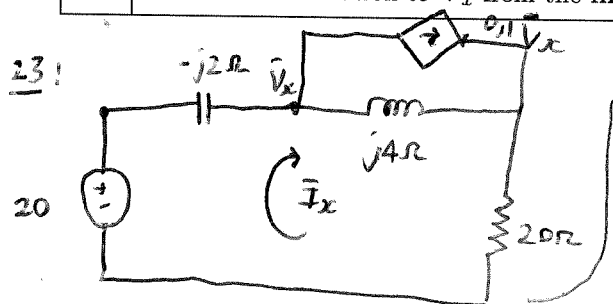
$$6\bar{\mathbf{I}}_1 + \bar{\mathbf{I}}_2[j10 + 4] = 20\angle 0^\circ + j10 + 6$$

$$6\bar{\mathbf{I}}_1 + \bar{\mathbf{I}}_2(4 + j10) = 26 + j10$$

$$\bar{\mathbf{g}} = 4 + j10 ; \bar{\mathbf{h}} = 26 + j10$$

Questions 23 to 24 [4]

Consider the circuit diagram below.

23 Find the contribution to V_x from the voltage source in **polar** form. [2]24 Find the contribution to V_x from the **independent** current source in **polar** form. [2]

$$\bar{V}_x = 20 + j2\bar{I}_x$$

$$\text{KVL: } -20 - j2\bar{I}_x + j4(\bar{I}_x - 0,1\bar{V}_x) + 20\bar{I}_x = 0$$

$$-20 - j2\bar{I}_x - j0,4[20 + j2\bar{I}_x] + j4\bar{I}_x + 20\bar{I}_x = 0$$

$$\bar{I}_x[-j2 + 0,8 + j4 + 20] = 20 + j8$$

$$\bar{I}_x = 1,031 \angle 16,31^\circ$$

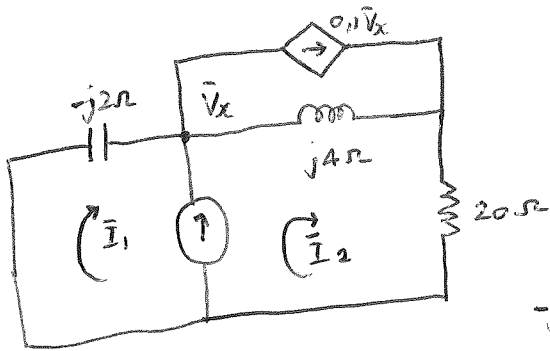
$$\bar{I}_x = 1,031 \angle 16,31^\circ$$

$$\bar{V}_x = 20 + j2(1,031 \angle 16,31^\circ)$$

$$= 19,92 \angle 5,82^\circ \text{ V}$$

24: P.T.O.

24:



$$\bar{I}_2 - \bar{I}_1 = j4 \Rightarrow \bar{I}_2 = \bar{I}_1 + j4 \quad (1)$$

$$\bar{V}_x = -(-j2)\bar{I}_1 \Rightarrow \bar{V}_x = j2\bar{I}_1 \quad (2)$$

$$-j2\bar{I}_1 + j4(\bar{I}_2 - 0,1\bar{V}_x) + 20\bar{I}_2 = 0 \quad (3)$$

$$(1)(2) \Rightarrow (3): -j2\bar{I}_1 + j4\bar{I}_2 - j0,4\bar{V}_x + 20\bar{I}_2 = 0$$

$$-j2\bar{I}_1 + j4(\bar{I}_1 + j4) - j0,4 \times j2\bar{I}_1 + 20(\bar{I}_1 + j4) = 0$$

$$j2\bar{I}_1 - 16 + 0,8\bar{I}_1 + 20\bar{I}_1 + j80 = 0$$

$$\Rightarrow (20,8 + j2)\bar{I}_1 = 16 - j80$$

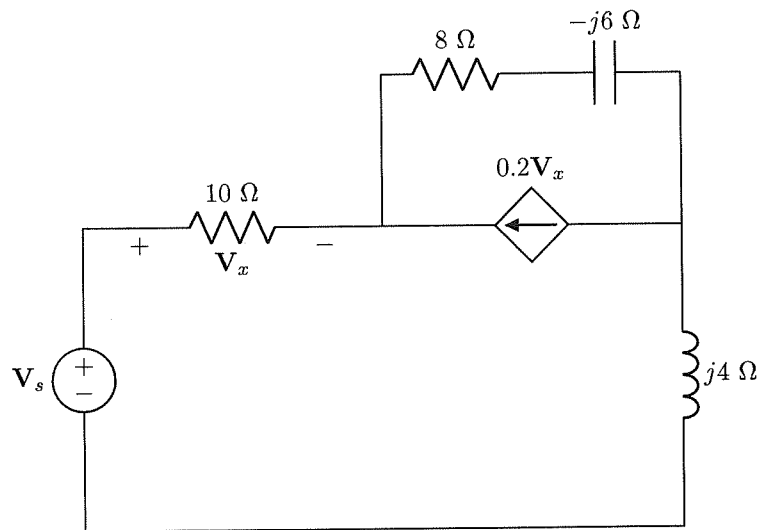
$$\Rightarrow \bar{I}_1 = \frac{16 - j80}{20,8 + j2}$$

$$\Rightarrow \bar{V}_x = j2\bar{I}_1 = 7,81 \angle 5,82^\circ \text{ V}$$

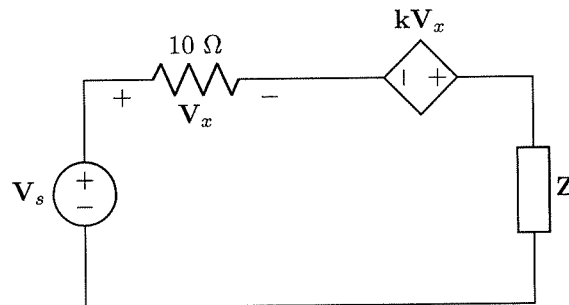
Questions 25 to 26 [4]

Consider circuits **A** and **B** below. Transform circuit **A** such that it is equivalent to circuit **B**.

Circuit A:



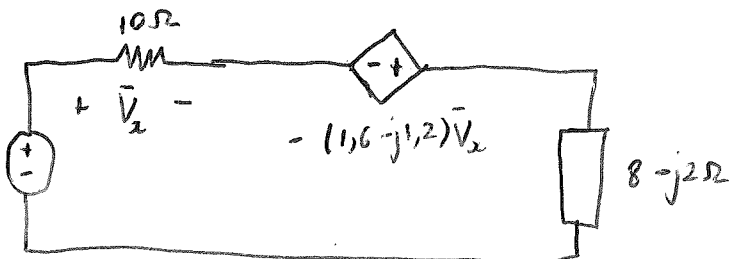
Circuit B:



25 Find **Z** in rectangular form. [2]

26 Find **k** in rectangular form. [2]

Turn over for additional space for Questions 25-26.

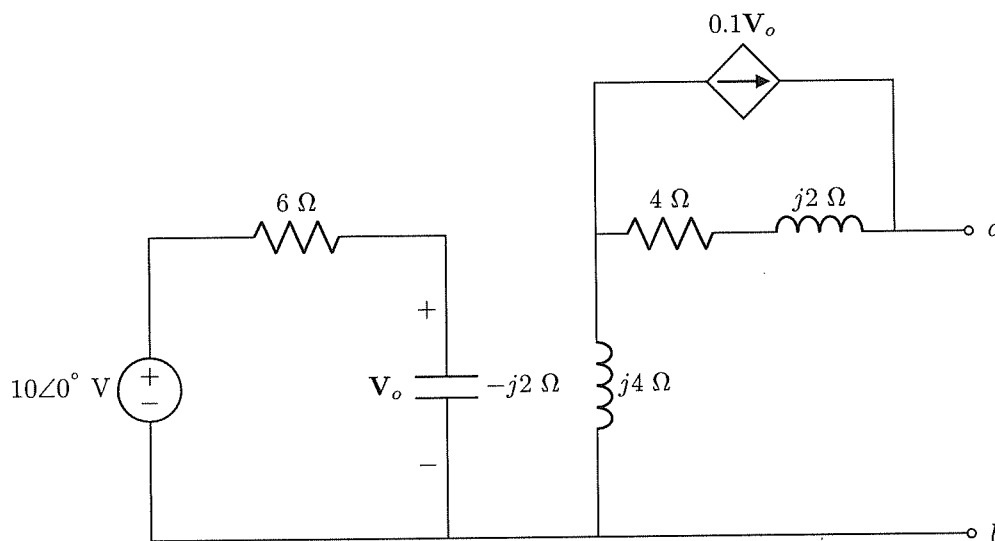


$$25: \hat{Z} = 8 - j2 \Omega$$

$$26: k = -1.6 + j1.2$$

Questions 27 to 28 [4]

Consider the circuit diagram below.

Find the Norton equivalent current I_N and impedance Z_N at terminals $a - b$ in rectangular form.27 Determine I_N . [2]28 Determine Z_N . [2]

27:
$$\bar{V}_o = 10\angle 0^\circ \left(\frac{-j2}{-j2 + 6} \right) = 1 - j3 \text{ V}$$

$$j4 \bar{I}_N + (4 + j2) [\bar{I}_N - 0,1(1 - j3)] = 0$$

$$\bar{I}_N [j4 + 4 + j2] = 0,1(1 - j3)(4 + j2)$$

$$\bar{I}_N = -0,038 - j0,192 \text{ A}$$

28:

$$\bar{I}_{sc} = \frac{1}{j2 + j4 + 4}$$

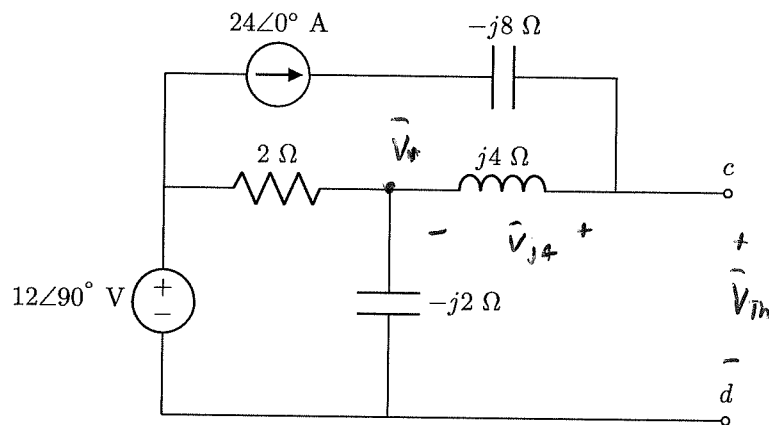
$$= \frac{-1}{13} + j \frac{3}{26}$$

$$\bar{Z}_N = \frac{1}{\bar{I}_{sc}}$$

$$= 4 + j6 \Omega$$

Questions 29 to 30 [4]

Consider the circuit diagram below.

Find the Thévenin equivalent voltage V_{Th} and impedance Z_{Th} at terminals $a - b$ in polar form.29 Determine V_{Th} . [2]30 Determine Z_{Th} . [2]

29:

$$\frac{\bar{V}_s - 12\angle 90^\circ}{2} + \frac{\bar{V}}{-j2} - 24\angle 0^\circ = 0$$

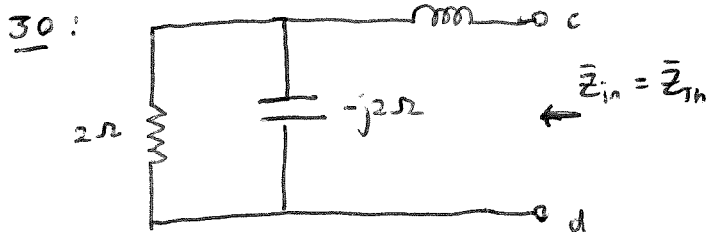
$$\bar{V} \left[\frac{1}{2} - \frac{1}{-j2} \right] = \frac{12\angle 90^\circ}{2} + 24$$

$$\bar{V} = 34,99 \angle -30,96^\circ \text{ V}$$

$$\bar{V}_{Th} = \bar{V} + \bar{V}_{j4}$$

$$= 34,99 \angle -30,96^\circ + 24 \angle 90^\circ$$

$$= 83,57 \angle 68,96^\circ \text{ V}$$



$$\bar{Z}_{Th} = \left(\frac{1}{2} + \frac{1}{-j2} \right)^{-1} + j4$$

$$= 3,162 \angle 71,565^\circ \Omega$$

PRACTICE ANSWER SHEET

Assessment ID	2	0	1	9	E	B	N	1	2	2	S	0	3
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	Answer	Unit
01	$L_{eq} = 48$	mH
02	$v_C = 0$	V
03	$i_L = 4$	A
04	$v_C(1 \text{ ms}) = 5$	V
05	$v_C(3 \text{ ms}) = 35$	V
06	Write 1, 2, or 3 in OCR form.	No units
07	$\phi_0 = 40$	°
08	$V = 10,44 \angle -35,50^\circ$	V
09	$Z = -55,5 + j206,07$	No units
10	$I_2 = 0,541 \angle 168,14^\circ$	No units
11	$V_s = 20,59 \angle 150,95^\circ$	V
12	$Z = 0,228 - j0,246$	Ω
13	$Z_{eq} = 1 - j2$	Ω
14	$Z = 8 - j6$	Ω
15	$a = 1$	No units
16	$b = -1$	No units
17	$c = -6 + j16$	No units
18	$d = 72$	No units
19	$e = -1$	No units
20	$f = -1,73 - j$	No units
21	$g = 4 + j10$	No units
22	$h = 26 + j10$	No units
23	Contribution to V_x from voltage source = $19,52 \angle 5,82^\circ$	V
24	Contribution to V_x from independent current source = $7,81 \angle 5,82^\circ$	V
25	$Z = 8 - j2$	Ω
26	$k = -1,6 + j1,2$	No units
27	$I_N = -0,038 - j0,192$	A
28	$Z_N = 4 + j6$	Ω
29	$V_{Th} = 83,57 \angle 68,96^\circ$	V
30	$Z_{Th} = 31,62 \angle 71,56^\circ$	Ω